Ниже представлен полный исправленный текст CONCEPT.md на английском языке с учётом всех принятых правок и рекомендаций рецензента.

FCLC — Federated Clinical Learning Cooperative

Privacy-Preserving Infrastructure for Medical AI Training Without Raw Patient Data Transfer

Version 6.1 — final (post-peer review) Date: April 7, 2026 Status: READY FOR SUBMISSION

One Sentence

We unite clinical and pharmaceutical data for AI training without transferring raw patient data, with measurable contributions from each participant and transparent benefit distribution.

The Problem

Data exists everywhere, but it is:

- In different formats — HIS, EHR, PACS, LIS, unstructured notes
- Under different legal constraints — GDPR, national laws, internal policies
- Nobody wants to share it — reputational, legal, and commercial risks

Result: Large medical AI models are trained on narrow datasets, while real clinical data remains unused. Doctors continue working without AI tools that could be built from already existing data.

The Solution

Each participant deploys a local node that:

1. Connects to HIS/EHR/PACS via adapters (HL7/FHIR, custom APIs)
2. De-identifies data with guarantees (identifier removal, quasi-identifiers, differential privacy)
3. Normalizes to a common schema (OMOP CDM)
4. Sends only de-identified aggregates or model updates (gradients, weights)

Central Orchestrator:

- Collects model updates

- Aggregates (federated averaging with secure aggregation)
- Maintains audit and version logs
- Calculates each participant's contribution

Data never leaves the clinic — only training signals do.

Privacy — Architectural, Not Rhetorical

Layer	Mechanism
1	Direct identifier removal (name, ID, address, exact date)
2	Quasi-identifier generalization (age groups, rare diagnoses → suppression)
3	Record-level re-identification risk assessment (k-anonymity, l-diversity)
4	Differential privacy (ε=2.0 per round, $\delta=10^{-5}$, total budget $\varepsilon_total \leq 10.0$, Gaussian mechanism, Rényi DP accounting)
5	Secure aggregation (orchestrator sees only aggregated result)

Result: Even if the orchestrator is compromised or traffic is intercepted, patient data cannot be reconstructed.

Threat Model and Security

Threat model:

- Orchestrator: honest-but-curious — follows protocol but may attempt to extract information from received data
- Nodes: up to 25% may be malicious (Byzantine) — sending incorrect updates to poison the model
- External attacker: may intercept network traffic

Security measures:

- Secure aggregation (SecAgg+): orchestrator does not see individual node updates. Current implementation uses LCG + DefaultHasher for architecture demonstration. Within this grant (WP3, budget €30k, months 3–6), we will replace this with ChaCha20+Poly1305 and Diffie-Hellman key agreement (Bonawitz et al., 2017) — a production solution already proven in MELLODDY.
- Differential privacy: sufficient noise to prevent data reconstruction from gradients
- Minimum batch size: ≥32 records to reduce gradient inversion attack risk (Zhu et al., 2019)

- Robust aggregation: Krum for Byzantine tolerance (up to 25% malicious nodes)
- Reputation scoring: nodes with anomalous behavior are automatically excluded from training

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Unified Clinical Schema

We use the OMOP Common Data Model (Observational Medical Outcomes Partnership) — the most widely adopted standard for observational research. Adaptation:

Domain	Standard
Diagnoses	ICD-10/11
Laboratory	LOINC
Medications	ATC, RxNorm
Procedures	SNOMED CT, CPT
Demographics	Generalized categories

The local node transforms data from HIS/EHR to OMOP on site. FHIR is supported as an alternative input format.

Measuring Contribution

Contribution is measured via approximated Shapley value (Federated Shapley Value).

For each participant, we compute their marginal contribution to global model performance. For scalability with 5–10 nodes, we use Monte Carlo sampling (100–200 iterations), providing acceptable accuracy with computational cost $O(n^2 \times M)$. For 10 nodes and 200 iterations, additional time is ~10 minutes per round — acceptable for asynchronous federated learning.

Advantages:

- Theoretically grounded: the only distribution scheme satisfying efficiency, symmetry, linearity, and dummy player axioms (Shapley, 1953)
- Manipulation-resistant: marginal contribution is difficult to inflate without genuine model improvement
- Proportional: participants receive credits strictly corresponding to their real impact on model quality
- Resistant to large participant dominance: Shapley value evaluates marginal contribution, not absolute data volume

Computational implementation:

- Shapley value estimation is performed by the central orchestrator after each round or epoch
- Monte Carlo method with 100–200 node permutation samples
- Scaling alternative: for >20 nodes, approximation via influence functions or stratified sampling

Participants receive contribution credits convertible to benefits.

Free-rider protection: Access to the global model and its new versions is granted only to participants whose accumulated contribution score exceeds a set threshold (e.g., 5% of the mean). This incentivizes active participation and prevents benefit without contribution.

What Participants Receive

Clinics:

- Access to best AI models without losing data control
- Licensing discounts
- Priority access to new versions
- Individual model fine-tuning on their own data
- Anonymous benchmarking

Pharmaceutical companies:

- Access to diverse real-world data without jurisdictional risks
- Post-marketing effectiveness analysis
- Accelerated R&D through federated cohorts

All participants:

- Proportional share of commercial revenue
- Participation in consortium governance

Governance Model

Legal Structure

The consortium is established as a non-profit partnership under the laws of the country of registration (Georgia or an EU member state). Participants sign a Consortium Agreement defining:

- Rights and obligations of participants
- Admission and exit procedures
- Decision-making processes
- Intellectual property distribution
- Liability and insurance

Central Orchestrator — a separate legal entity (consortium subsidiary) operating under contract.

Two-Tier Model for Different Participant Types

Given the fundamentally different interests and risk profiles of clinics and pharmaceutical companies, a two-tier structure is introduced:

Tier	Participants	Rights	Obligations
Clinical Tier	Hospitals, diagnostic centers, clinics	Full voting on clinical matters; model access; individual fine-tuning	Provision of de-identified clinical data; ethical compliance
Industrial Tier	Pharmaceutical companies, CROs	Observer status on clinical matters; full voting on commercial matters; access to aggregated data	Provision of trial data; financial support (Phases 2–3)

Multi-Stakeholder Board

Category	Status
Clinic representatives (3–5)	Voting
Pharmaceutical representatives (2–3)	Voting
Patient organizations (1–2)	Observers
Data protection lawyers (1)	Experts
Ethics committees (1)	Experts
Technical auditors (1)	Non-voting

External Clinical Advisory Board

To ensure independent clinical expertise, an external advisory board is established:

Role	Functions	Budget
2–3 independent clinicians	Clinical significance assessment, approach validation, improvement recommendations	€20,000 (honoraria, travel)

Board meetings are held twice a year online.

Six Governance Principles

1. Raw data never leaves the clinic — architectural requirement, auditable
2. Every outgoing contribution is privacy-verified — automatically and selectively manually
3. Every contribution is weighted by utility — via Shapley value system
4. Every participant receives a contribution score — transparent, with appeal possible
5. Every model is audited for bias — mandatory before release
6. Commercial benefits are distributed transparently — via board-approved formula

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Legal Framework

Data Use Agreement (DUA)

Standard DUA template signed by each participant includes:

- Definition of data scope and type
- Prohibition on raw data transfer to third parties
- Obligation to comply with de-identification protocols
- Liability for data breaches
- Termination procedures

Process: DUA negotiations begin 3 months before technical development. Standard template approved by consortium lawyers. DUA signing is a condition for technical integration.

Ethical Approval (IRB)

Each participating institution obtains local IRB approval before participation. The process includes:

- Confirmation that only de-identified data is transferred
- Patient informed consent (where required by national law)

- Annual re-approval

Strategy: IRB approval is obtained in parallel across all participating clinics, with total duration up to 3 months. The coordinator (WP4) provides document templates and supports institutions through the process.

Support: The Medical Consultant provides document templates, advises clinics on ethical approval, and accompanies the IRB process.

Data Steward

Each participating organization appoints a Data Steward responsible for:

- Coordinating data provision per DUA
- Ensuring compliance with de-identification protocols
- Liaising with local IRB
- Reporting any data-related incidents

GDPR (EU)

- Applicability: if EU institutions participate or EU citizen data is processed
- Article 9 (special categories of data): model updates are considered health data processing
- Legal basis: patient informed consent or national legal provision for secondary data use (GDPR Art. 9(2)(i) and (j))
- DPIA: mandatory for each EU participant
- Breach notification: within 72 hours

Georgian Personal Data Protection Law (PDPL, 2011, amended 2023)

- Compliance: the law is harmonized with GDPR; requirements are similar
- Consent: required for medical data processing (unless otherwise provided by law)
- Breach notification: within 72 hours to the Personal Data Protection Service of Georgia
- National authority: Personal Data Protection Service of Georgia
- Interpretation note: In case of interpretive divergence, the consortium follows the stricter standard (GDPR + EDPB Guidelines 05/2014). The admissibility of the de-identification method will be confirmed by a preliminary opinion from the Georgian regulator (requested in April 2026).

Intellectual Property and Financial Model

IP Regime

Result Type	Rights
Base model	Joint consortium ownership, licensed to participants at preferential rate
Participant adaptations (fine-tuning)	Exclusive participant ownership
Local components (adapters, scripts)	Participant ownership with open license to consortium
Methodology, publications	Open Access with consortium attribution

Financial Model

Phase	Period	Mechanism
Phase 1 — Pilot	12 months	Grant funding (Horizon Europe), participants pay nothing, accumulate credits
Phase 2 — Scaling	Launches with ≥10 participants and validated model	Entry fees for new participants; licensing to external organizations; 70% revenue to participants proportional to credits (Shapley value), 30% to platform development
Phase 3 — Sustainability	After operational stability (≈36 months)	Annual membership fees (differentiated by participant type); licensing as primary income; custom training services

Membership fees (Phase 3):

- Clinics: from €5,000/year (depending on institution size)
- Pharmaceutical companies: from €50,000/year
- Research organizations: €0–5,000/year

Model licensing (for external organizations):

- Standard license: from €100,000 per model
- Custom adaptations: by negotiation

Free-rider protection mechanism: Access to the global model is granted only to participants whose accumulated contribution score (Shapley value) over the last 12 months exceeds 5% of the consortium mean. Participants below the threshold lose access until activity resumes.

MVP (Minimum Viable Product)

Scenario

Prediction of hospitalization risk in type 2 diabetes patients within 12 months

Justification

- Type 2 diabetes is widespread; data exists in most clinics
- Outcome (hospitalization) is clearly defined and recorded
- High clinical value — cost reduction, improved outpatient management
- No complex imaging required (only structured data)

Pilot Participants

Type	Count	Candidates (status)
Clinics	3–5	Aversi Clinic (negotiations started), GeoHospitals (preliminary agreement), Iashvili Children's Hospital (diabetology department, negotiations started)
Pharmaceutical company	1	TBC (active search)
EU technical partner	1	TBC (DFKI, Fraunhofer, or EU university)
EU medical partner (optional)	0–1	TBC (Charité Berlin, Karolinska Institutet, Erasmus MC) — to strengthen institutional base

Key Experts (hired with grant funds)

Role	Functions	FTE	Budget (€)
Medical Consultant	Clinical validation, physician liaison, result interpretation, IRB preparation	0.5	60,000
Technical Expert (Database Systems)	ETL development, PostgreSQL/OMOP setup, HIS/EHR integration	1.0	60,000

Realistic Timeline

Phase	Duration	Description
Legal preparation	3 months	DUA preparation, IRB approval at each clinic, PDPL/GDPR legal memorandum, Data Steward appointment
Technical development	3 months	Local node development, normalization, Shapley value scoring, aggregation

Deployment and pilot	6 months	Deployment at 3–5 clinics, federated learning, validation
Total	12 months	

Minimum Data Set

Field	Source	Format
Age	Demographics	5-year intervals
Sex	Demographics	M/F/Other
Type 2 diabetes diagnosis date	Diagnoses	Year (no day)
HbA1c (most recent)	Laboratory	Numeric
BMI	Physical exam	Numeric
Complications (nephropathy, retinopathy)	Diagnoses	Binary flags
Hospitalization in last 12 months	Events	Binary flag
Hospitalization in next 12 months (outcome)	Events	Binary flag (target)

Success Criteria

- Model achieves AUC > 0.75 on held-out validation set
- No raw patient record leaves any institution (audit-confirmed)
- Shapley value scoring works and is accepted by participants
- Technical interoperability ensured for all 3–5 clinics
- All legal requirements (DUA, IRB) completed for each clinic

Clinical Validation

Clinical validation is ensured through:

1. Medical Consultant (0.5 FTE) — independent physician-researcher with diabetology experience, responsible for:
    - Verification of clinical significance
    - Liaison with physicians at pilot clinics
    - Result interpretation
    - IRB materials preparation

2. Pilot clinic physician involvement — on-site validation participation, clinical applicability feedback
3. External Clinical Advisory Board — 2–3 independent clinicians for clinical significance and methodology assessment

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Technical Stack

Component	Choice	Justification
Federated learning	Flower + OpenFL	Flower — flexibility, OpenFL — maturity for medical applications
Aggregation algorithm	FedProx (Li et al., 2020, MLSys) with μ=0.1–1.0	Robustness to non-IID data — critical for heterogeneous clinical data; μ tuned during pilot
Backup algorithm	SCAFFOLD (Karimireddy et al., 2020)	For high heterogeneity cases; switching based on convergence monitoring
Robust aggregation	Krum (Blanchard et al., 2017)	Tolerance to up to 25% malicious nodes; chosen for MVP for balance of effectiveness and complexity
Secure aggregation	SecAgg+ (Bonawitz et al., 2017) — full implementation in WP3	Orchestrator does not see individual updates
Differential privacy	TensorFlow Privacy / Opacus with Rényi DP	Standard libraries; $\varepsilon=2.0$/round, $\delta=10^{-5}$, total budget $\varepsilon_total\leq10.0$
Contribution scoring	Federated Shapley Value (Wang et al., 2020) with Monte Carlo approximation (M=100–200)	Theoretically grounded, manipulation-resistant; computational cost ~10 min/round for 10 nodes
OMOP normalization	OHDSI WhiteRabbit + custom ETL	Proven tooling
Interoperability	HL7/FHIR → OMOP	FHIR for exchange, OMOP for analysis
Node storage	PostgreSQL + OMOP CDM	Open source, supports required volumes
Security	TLS 1.3, disk encryption, HSM for keys	Medical data compliance

Node Hardware Requirements (Minimum)

Component	Minimum Requirements
CPU	8 cores, 3.0+ GHz
RAM	32 GB
GPU	NVIDIA Tesla T4 or equivalent with 16 GB VRAM (for training)
Storage	1 TB SSD (for OMOP CDM and logs)
Network	100 Mbps, static IP
Security	TPM 2.0 or HSM for encryption keys

Comparison with Existing Solutions

Solution	Description	FCLC Distinction
FedAvg (McMahan et al., 2017)	Basic federated averaging algorithm	Use FedProx/SCAFFOLD for non-IID data
FATE (WeBank)	Full FL framework deployed in banking and healthcare in China	Cooperative (non-profit) governance model + Shapley value incentives
NVFlare (NVIDIA)	Used in 20-institution COVID-19 study (Rieke et al., 2020)	Can use NVFlare as technical substrate, adding governance layer
OpenFL (Intel)	Production-grade FL for medical applications	Same as NVFlare — technical substrate
FeTS (Pati et al., 2022)	Largest FL study in oncology (71 institutions, 6 continents)	Convergence results on non-IID data relevant to FCLC MVP
MELLODDY (Bender et al., 2021)	Consortium of 10 pharma companies for FL in drug discovery	Closest precedent to proposed model with pharmaceutical nodes; adds cooperative structure and transparent benefit distribution
PySyft / OpenMined	Cryptographic primitives libraries	Used as technical substrate
Flower (flwr)	Framework-agnostic FL library	Used as foundation with governance layer on top

FCLC uniqueness:

- Focus on low-resource national health systems (Georgia, South Caucasus region)
- Cooperative (non-profit) governance model with two-tier structure (clinics + pharma)
- Theoretically grounded incentive system (Shapley value) instead of ad hoc formulas
- Integration with AIM ecosystem (DeepSeek API, unified data governance policies)
- Flexible consortium structure with hired experts instead of institutional partners, reducing administrative barriers

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Unified Scientific Ecosystem: Ze · CDATA · BioSense · FCLC

Overview

FCLC is not a standalone platform — it is the validation layer of a unified scientific ecosystem spanning four interconnected projects. Each project solves a distinct problem, but together they form a closed scientific loop: from theory to measurement, from prediction to population-scale validation.

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│                                                AIM ECOSYSTEM                                                                 │
│                                                                                                                              │
│  Ze Theory        CDATA           BioSense        FCLC                                                                       │
│  ──────────────   ──────────────   ──────────────   ──────────────                                                          │
│  Theoretical  →  Mechanistic  →  Measurement  →  Population                                                                  │
│  foundation       aging model      tool            validation                                                                 │
│                                                                                                                              │
│  τ_Z, v*, χ_Ze    dD/dt, R²=0.84   χ_Ze(EEG/HRV/    AUC=0.742                                                                 │
│  T/S events       32 parameters    VOC), AUC=0.715  N clinics                                                                 │
│                                                nRF52840 Rust                                                                 │
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Synergy Matrix

Project	Provides to others	Receives from others
Ze Theory	Mathematical language (v*, χ_Ze, τ_Z, T/S); unified interpretive framework	Empirical validation via BioSense; population data via FCLC
CDATA	Mechanistic D(t) model; 5 falsifiable predictions; MCMC parameters	BioSense as D(t) measurement tool; FCLC for multi-center validation
BioSense	Continuous 24/7 data stream; χ_Ze(EEG/HRV/VOC); FCLC node on each device	Ze Theory as χ_Ze justification; CDATA as interpretive framework for χ_Ze decline
FCLC	Population-scale validation; statistical power; privacy	BioSense as data source; CDATA as prediction model; Ze as theory

Publications (as of April 2026)

Paper	Repository	Target Journal	Status
Ze Theory (quantum mechanics)	Ze-public	Longevity Horizon	Published, DOI 10.65649/a874t352
CDATA	CDATA-public	Annals of Rejuvenation Science	Published, DOI 10.65649/cynx718
Ze observation as τ_Z expenditure	Ze-public / Articles	Physical Review Letters / Found. Phys.	Preprint, ready for submission
Ze–CDATA Bridge Equations	Ze-public / Articles	Physical Biology	Preprint, ready for submission
BioSense flagship paper	BioSense-public / Articles	npj Digital Medicine (IF ≈ 15)	Preprint, ready for submission
FCLC federated protocol	FCLC-public	Nature Machine Intelligence	In preparation

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Technology Readiness Levels (TRL)

Component	Current TRL	Target TRL	Justification
Privacy stack (excluding SecAgg)	TRL 4	TRL 6	Validated on MIMIC-IV
SecAgg (production)	TRL 2	TRL 5	Requires crypto replacement (WP3)

FedProx + Krum	TRL 4	TRL 6	Validated, batches ≥32
Shapley value scoring	TRL 4	TRL 6	Works for 5 nodes
HIS/EHR integration	TRL 3	TRL 5	Requires clinic pilot

Validation Results (MIMIC-IV, added 2026-04-06)

Paper JMIR Medical Informatics (v3.0, 2026-04-05) contains first quantitative validation of FCLC v1.0.

Scenario

- Dataset: MIMIC-IV, N=12,543 T2D patients, 47,892 visits
- Task: Predict 30-day readmission (binary classification)
- Model: Logistic regression, 15 features (age, sex, LOS, prior hospitalizations, HbA1c, glucose, creatinine, BP, HR, temp, 5 comorbidities)
- Nodes: 5 non-overlapping partitions (2,108–2,892 patients each)

Performance Results (AUC-ROC)

Scenario	AUC (with DP)	AUC (without DP)	Local-only	Improvement
Baseline (5 honest nodes)	0.742±0.021	0.763±0.019	0.712±0.031	+3.0 pp (4.2%, p=0.008)
Free-rider (4 honest + 1 noise)	0.724±0.026	0.745±0.024	0.708±0.033	−2.4% vs baseline
Byzantine (4 honest + 1 attack ×100)	0.735±0.023	0.758±0.021	0.710±0.030	+3.5%

Privacy–Utility Trade-off (ε vs AUC)

ε per round	ε_total (5 rounds)	AUC	AUC loss
∞ (no DP)	∞	0.763	—
4.0	20.0	0.758	−0.7%
2.0	10.0	0.742	−2.8% ← chosen
1.0	5.0	0.718	−5.9%

0.5	2.5	0.685	−10.2%

Privacy Audit Metrics

Metric	Result	Interpretation
k-anonymity	97.3% (2.7% suppressed)	✅
Gradient inversion PSNR	<18 dB	✅ Reconstruction impossible (threshold ~20 dB)
SSIM	0.12±0.04	✅ No structural similarity
Cosine similarity	0.09±0.03	✅ Orthogonal vectors
Attack success	0/32 (0%)	✅

Shapley Scores (baseline scenario)

Node	Shapley score	Rank
A	0.342±0.021	1
B	0.318±0.019	2
C	0.309±0.022	3
D	0.298±0.018	4
E	0.287±0.020	5

Correlation with local AUC: r=0.87, p=0.05. Free-rider: 0.089 < threshold 0.0975 → correctly detected (p<0.001).

Runtime

Component	Time
Full round	47±8 sec
Shapley computation (M=150)	12.3±2.1 sec
Data de-identification	8.4±1.2 sec
Local training	2.1±0.3 sec
Network transfer	2.3 KB/round

Funding Instrument Selection

Recommended instrument: EIC Pathfinder Open

Criterion	Match
Breakthrough innovation	Combination of federated learning + secure aggregation + Shapley value incentives + two-tier cooperative governance — unique
TRL	2–4 (current), post-pilot (12 months) — TRL 4
Budget	Up to €4M — justified by development, deployment, legal preparation, coordination
Consortium	Georgia (associated country) + minimum 2 EU partners (in formation)
Deadline	May 12, 2026

Consortium Structure

Partner	Country	Type	Role	Status
Phasis Academy (J. Tkemaladze)	Georgia	Research	PI, Coordinator	✅ Confirmed
Giorgi Tsomaia	Georgia	Independent expert	Co-Investigator, Lead of WP2 and WP4	✅ Confirmed
Medical Consultant	Georgia/EU	Clinical	Clinical validation, IRB, physician liaison	Vacant (in budget)
Technical Expert (Database Systems)	Georgia/EU	Technical	ETL development, OMOP, PostgreSQL, HIS integration	Vacant (in budget)
Aversi Clinic	Georgia	Clinical	Pilot site	Negotiations started
GeoHospitals	Georgia	Clinical	Pilot site	Preliminary agreement
Iashvili Children's Hospital	Georgia	Clinical	Pilot site	Negotiations started
EU Technical Partner — TBC	EU	Technical	FL infrastructure, secure aggregation	Active search

Potential EU Partners (as of April 2026)

Partner	Country	Contact status	Probability
DFKI (German Research Center for AI)	Germany	Inquiry sent April 1, 2026	Medium

Fraunhofer IAIS	Germany	Planned for week 15	Low/Medium
Saarland University (FL Lab)	Germany	Preliminary discussions	Medium
Karolinska Institutet	Sweden	Backup option	Low

Plan B: If no EU partner is confirmed by April 20, 2026, the application will be submitted with one Georgian technical partner plus an explanation that EU coordination will be ensured through Horizon Europe guarantees.

Budget Estimate

Category	Estimate (€)
Legal preparation (DUA, IRB, memorandum, Data Steward)	55,000
Medical Consultant (0.5 FTE, 12 months)	60,000
Technical Expert (Database Systems) (1.0 FTE, 12 months)	60,000
External Clinical Advisory Board	20,000
Local node development (3–4 FTE, 12 months)	280,000
Normalization and ETL (2 FTE, 12 months)	120,000
Shapley value scoring and aggregation (2 FTE, 24 months)	250,000
Federated learning and security (2 FTE, 24 months)	250,000
WP3 — SecAgg production crypto (ChaCha20+DH)	30,000
Pilot clinic deployment (3–5)	100,000
Coordination and management (2 FTE, 36 months)	320,000
WP4 (Governance)	220,000
Dissemination, communication, events	130,000
Equipment, infrastructure	120,000
Travel	100,000
Contingency (10%)	210,000
Total	2,275,000

Note: EIC Pathfinder allows budget up to €4M. The above estimate is baseline; with expanded consortium and scope, the budget may increase to €3–3.5M.

Risk Matrix

Risk	Probability	Impact	Mitigation	Owner
Re-identification	Low	Critical	Multi-layer privacy stack + audit	Privacy Lead
Gradient inversion	Medium	Critical	DP noise, min batch ≥32, robust aggregation	Technical Lead
Data poisoning	Medium	High	Krum/median aggregation, reputation scoring, outlier detection	Technical Lead
Bias	High	Medium	Distribution monitoring, stratified validation	Governance Board
Label inconsistency	High	Medium	Quality scoring, clinician feedback, Medical Consultant	Clinical Lead
Large participant dominance	Medium	Medium	Shapley value ensures fair marginal contribution assessment	Governance Board
DUA delays	High	High	Start process 3 months before technical development; standard template; parallel negotiations	Legal Lead
IRB delays	High	High	Parallel approval across clinics; Medical Consultant support; document templates	Legal Lead
Participant dropout	Medium	Medium	Waiting list; backup participants; modular architecture allows flexible node connection/disconnection	Coordinator
No medical Co-PI	Medium	High	Medical Consultant (0.5 FTE) + External Clinical Advisory Board	Coordinator
Expert hiring delays	Medium	Medium	Start search at application stage; backup candidates	Coordinator
Regulatory distrust	Medium	Critical	Engage regulators at pilot stage; legal memorandum; transparent documentation	Legal Lead
Clinician distrust	High	Medium	Explainable models (SHAP, LIME); physician involvement in validation; pilot data demonstration	Clinical Lead

HIS incompatibility	Medium	High	Modular adapters; FHIR support; file upload fallback; pre-audit infrastructure	Technical Lead
No EU partner	Medium	Critical	Active search via DFKI, CLAIRE networks; letters of support by April 2026; Plan B documented	Coordinator

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Roadmap v2.0 (Post-Pilot)

Limitations of v1.0 to be addressed in the next version:

v1.0 Limitation	v2.0 Solution	Priority
Linear DP accounting (conservative)	Rényi DP — tighter bounds, subsampling support → less AUC loss at same ε	High
Orchestrator honest-but-curious	SecAgg+ (full) — cryptographic masking; orchestrator sees only aggregate	High
Synchronous rounds	Asynchronous rounds — not all nodes must participate simultaneously	Medium
Logistic regression only	Transformer validation — deep learning with DP (feasibility study in 2026)	Medium
Shapley slow for >20 nodes	Influence functions — Shapley approximation for >20 nodes	Medium
GDPR only mentioned	GDPR Art. 22 analysis — "right to explanation" compliance via Shapley attribution	Low

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Next Steps

Deadline	Task	Status
Week 1–2	Start recruiting Medical Consultant and Technical Expert; prepare job descriptions	Planned
Week 1–2	Start DUA and IRB process for pilot clinics (Aversi, GeoHospitals, Iashvili)	Launching
Week 2–3	Prepare legal memorandum on Georgian PDPL and GDPR	Planned
Week 2–3	Send inquiries to EU technical partners (DFKI, Fraunhofer, Saarland University)	Planned
Week 3–4	Prepare letters of support from clinics	Planned
Week 4–5	Prepare Part A and Part B of application	Planned

By May 12, 2026	Submit application

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